

Northern Humboldt model validation

Field	Entry
Region	Humboldt Current, LME_013. Selected-model evaluation of the northern Peruvian food web. Regional workbook
Catch source	Sea Around Us, 1950–2019; confidence reference: 2019 landings. All 218 exact labels, including zero catch and unidentified categories. Catch and classic inputs Reference arithmetic
Selected article	Chiaverano et al. (2018). Evaluating the role of large jellyfish and forage fishes as energy pathways, and their interplay with fisheries, in the Northern Humboldt Current System. Progress in Oceanography 164, 28–36. DOI 10.1016/j.pocean.2018.04.009. Source PDF DOI
Other known articles	Neira et al. (2026), Chilean Patagonia 1980–2020 food-web modelling, DOI 10.1016/j.pocean.2025.103631. Its static 1980 model is retained as previous-selection evidence. Tam et al. (2008), Northern Humboldt trophic linkages under La Niña and El Niño, is a predecessor; its stocks cannot replace this model without a verified crosswalk. Selected-model review
Selected model	Chiaverano2018_detailed_Northern_Humboldt_1995_1998; selected audited detailed Northern Humboldt baseline, 1995–1998. 39 ecological stocks plus two source fleet nodes, over 165,000 km ² at 4°S–16°S and up to 111 km offshore. The paper counts 36 living groups and three detritus pools. Anchovy eggs (36) are a nonfeeding pool in the supplement, giving four operational detritus pools in this translation. The runtime adds one synthetic diet_import source. Published source representation Corrected model JSON
Other models from the same article	The aggregated baseline has 24 ecological groups plus one pooled fishery node (25 source nodes), over the same area and period. It merges source groups and uses rounded printed values. Full detailed detritus/fleet routing is retained only in the resolved version. The four structural scenarios alter jellyfish consumption, forage-fish consumption or fishing rates; they are separate scenarios, not replacement static baselines.
Selection rationale	Article - Requested source review to test the local source evidence and its applicability to Northern Humboldt catches. Model - Prioritize the resolved supplement because its distinct taxa, hake stages, two fleets and detailed diets/routing permit a more faithful audit than the aggregated alternative. The user selected this audited model after auto returned finite, nonnegative SPPR in all three methods. Selection does not establish scientific validity or researcher validation.

Field	Entry
Model extraction	Accepted corrections: Chrysaora plocamia (7) diet is 68% mesozooplankton, 17% macrozooplankton and 15% anchovy eggs, with zero direct diatom/small-gelatinous links. Author Table H I14/J14 reconstructs the 68/17 fractions; F14/K14 are zero and Table B H39 retains the egg fraction. This resolves the contradictory Table B diet totaling 1.045 without changing small-gelatinous biomass. Sardine (9) landings follow the article's explicit 1.4 t/km²/year correction; reported discards 0.17462648325 remain. Total JSON export is 1.57462648325; estimated EE is recalculated to 0.68720654. Article Table 1 and Scenario III still support the older high-catch branch. These are documented reconstructions/derived corrections, not newly printed author values. The corrected model passes mass balance with living BA=0 and normalization compensation disabled. GS retains source assimilation efficiencies. Runtime normalization of the remaining diets is separate; pool EE/BA and detritus-fate identity replacement remain computational transformations. Native fleet discard-return ancestry is unsupported in the SPPR translation. Published-source extraction audit Source conflicts Prior loader matrix audit Retained departures and evidence limits:model notes
GE diagnostics	FAIL: unpooled detritus recycling diverges. Pooled coefficients are nonnegative and budget-balanced, but the overall diagnostic retains the unpooled divergence failure. Unpooled negative SPPR sources: Anchovy eggs (36), Pelagic detritus (38) and Benthic detritus (39). Every affected recipient, including unfished groups, is listed separately under its source in the linked Excel attachment. rho_living = 0.7000; unpooled b = 1.6356. Unpooled Detritus SPPR: Anchovy eggs (36) = -2.8590; Fishery offal (37) = 0.0000; Pelagic detritus (38) = -0.9310; Benthic detritus (39) = -4.3110. Pooled b = 0.7459. No negative pooled SPPR entries across any basal-source column or recipient. Pooled Detritus SPPR: Anchovy eggs (36) = 5.4726; Fishery offal (37) = 5.4726; Pelagic detritus (38) = 5.4726; Benthic detritus (39) = 5.4726. Strict model balance and SPPR budget balance pass in both configurations. Full GE return Labeled GE matrix Complete negative source-to-group pairings (Excel)
TE diagnostics	FAIL: primary-production budget does not close despite the corrected model passing mass balance. No negative SPPR entries across any basal-source column or recipient, including unfished groups. rho_living = 0.7316. Detritus SPPR: Anchovy eggs (36) = 0.0000; Fishery offal (37) = 0.0000; Pelagic detritus (38) = 0.3043; Benthic detritus (39) = 0.0000. PP budget gap = 14%. Five EE=0 living groups and very small jellyfish EE remain formulation concerns. The engine has no pooled-detritus scaling for TE: explicitly requesting pooling returns the same TE matrix and gap. Full TE return Labeled TE matrix
SPPR calculation	[Researcher calculation choices and notes]

Field	Entry
Geographic fit	Approximate A. Region covered by study: 6.0–8.0%. Approximate B. Study covered by region: 90–100%. The northern shelf domain leaves most of this LME outside the model. Boundary assessment
Temporal fit	Static 1995–1998 baseline applied to catch years 1950–2019; 2019 is 21–24 years later. Fixed coefficients and source catch/biomass allocations do not represent annual ecosystem or catch-composition changes. Publication year 2018 is separate from model period.
Other	Both unpooled (never) and pooled (always) configurations were run on the corrected canonical model. Living BA remains zero; remaining source diets are normalized at runtime. With Egestion remains FAIL overall: rho_living = 0.5600; unpooled b = 1.1198; pooled b = 0.8554. Unpooled negative sources are Anchovy eggs (36), Pelagic detritus (38) and Benthic detritus (39); complete recipient pairings are in the linked attachment. Unpooled Detritus SPPR: Anchovy eggs (36) = -21.3065; Fishery offal (37) = 0.0000; Pelagic detritus (38) = -14.7602; Benthic detritus (39) = -21.0164. Pooled Detritus SPPR: Anchovy eggs (36) = 10.7629; Fishery offal (37) = 10.7629; Pelagic detritus (38) = 10.7629; Benthic detritus (39) = 10.7629. Pooled With Egestion has no negative entries and passes SPPR budget balance, while detritus SPPR exceeds 10 and the overall unpooled-divergence failure remains. Taxon allocations retain their documented published native-landings proxies; the sardine correction does not silently alter those assumptions. Updated regional estimates remain provisional and production-ineligible. Map refresh awaits researcher review. Full direct report Updated workbook calculations Complete negative source-to-group pairings (Excel)
Open issues and next action	
Review and reproducibility	Researcher name: [name] review date: [dd/mm/yyyy]

Taxon mapping and coverage

Reference: 2019 landings; all recorded labels, all model groups and no taxon exclusions.

39 source ecological groups plus two fleet records and one computational import; 218 catch taxa.

Simple-chain PPR totals 277,099,551.43 t C in 2019.

34 labels lack classic TL/coefficient inputs; all have zero 2019 landings and contribute zero annual PPR. Their coefficients remain unavailable.

Method: independent simple trophic chain using the saved regional coefficients.

[Excel taxon appendix](#) | [Descriptive Sources sheet](#)

Overall confidence	Taxa (n)	Catch (%)	Simple-chain PPR (%)
High	70	87%	71%
Medium	104	9.4%	24%
Low	17	0.24%	0.26%
Very low	27	3.2%	4.9%
Unresolved	0	0%	0%

Overall confidence uses the weaker required membership or allocation component.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Explicit inclusion of the taxon in the source model	High	71%
Supported extension from the source's listed representative taxa	Medium	7.3%
Source membership connected through a verified synonym or documented spelling correction	High	6.7%
Supported ecological assignment requiring a group-boundary interpretation	Medium	6.6%
Closest represented ecological or taxonomic analogue	Very low	3.3%
Assumed eligible group set supported by taxonomy and ecology	Medium	3.2%
Approximation of a broad reporting category across plausible model pools	Very low	1.6%
Partial or competing ecological fit	Low	0.26%
Unambiguous fit to a documented taxonomic or ecological group	High	0.15%

Membership evidence: exact supplement group definitions, source taxonomic lists and authoritative synonym/habitat records. [Complete mapping evidence](#)

Allocation weight rules

Plain-language rule	Confidence	PPR percentage
One group; no allocation split required	High	86%
Allocation proportional to native source-model landings	Medium	14%
Complete source-model biomass fallback after unusable landings	Medium	0.046%

Weights use measured composition only when applicable. The source-model landings proxy is used before biomass; assumed proportions are fixed across catch years, with stage, fishery and geographic transfers recorded. [Allocation evidence](#)

Each percentage is the independent simple-chain PPR associated with taxa using that rule.

Unresolved taxa: none. 27 Very low decisions account for 3.2% of catch and 4.9% of independent simple-chain PPR.

Very low decisions

Affected taxa	Why confidence is very low
Actinopterygii; Gadiformes; Marine finfishes not identified; Marine fishes not identified; Marine groundfishes not identified; Marine pelagic fishes not identified; Perciformes; Scorpaeniformes	Broad fish reporting labels have unknown composition across named and residual pools, historical taxonomic scope, and potentially unrepresented deep-water members; native landings approximate their mixtures.
Batoidea; Chondrichthyes; Elasmobranchii; Myliobatidae	Broad/historical cartilaginous labels cross benthic and water-column pools with unrepresented chimaera or filter-feeding ray components.
Callorhynchus callorynchus	Benthic chimaera uses an elasmobranch analogue across the Holocephali/Elasmobranchii boundary.
Decapoda; Malacostraca; Miscellaneous marine crustaceans	Benthic/pelagic mixture uses macrozooplankton/benthos biomass because native landings are zero; a 2–20 mm size mismatch remains.
Dissostichus eleginoides; Hoplostethus atlanticus	Deep, cold-water fishes use a medium-demersal analogue with population, depth and size mismatches.
Galatheidae; Grimothea monodon	Squat lobsters lack a source pool; macrozooplankton/benthos biomass weights retain the 2–20 mm size mismatch.
Galaxias maculatus	Amphidromous galaxiid lacks a corresponding source population; a coastal small-fish analogue is assumed.
Macruronus magellanicus; Merluccius australis; Merluccius gayi; Micromesistius australis	Non-Peruvian hake/cod-like fishes use Peruvian hake groups; species, southern ecology and size boundaries differ, and native stage landings are transferred.
Mobula birostris	Pelagic filter-feeding manta uses the closest chondrichthyan pool despite the source apex-predator correspondence.
Ophichthus	Snake eel mapped to unnamed conger pool despite taxonomic identity differing from the regional congrio interpretation.

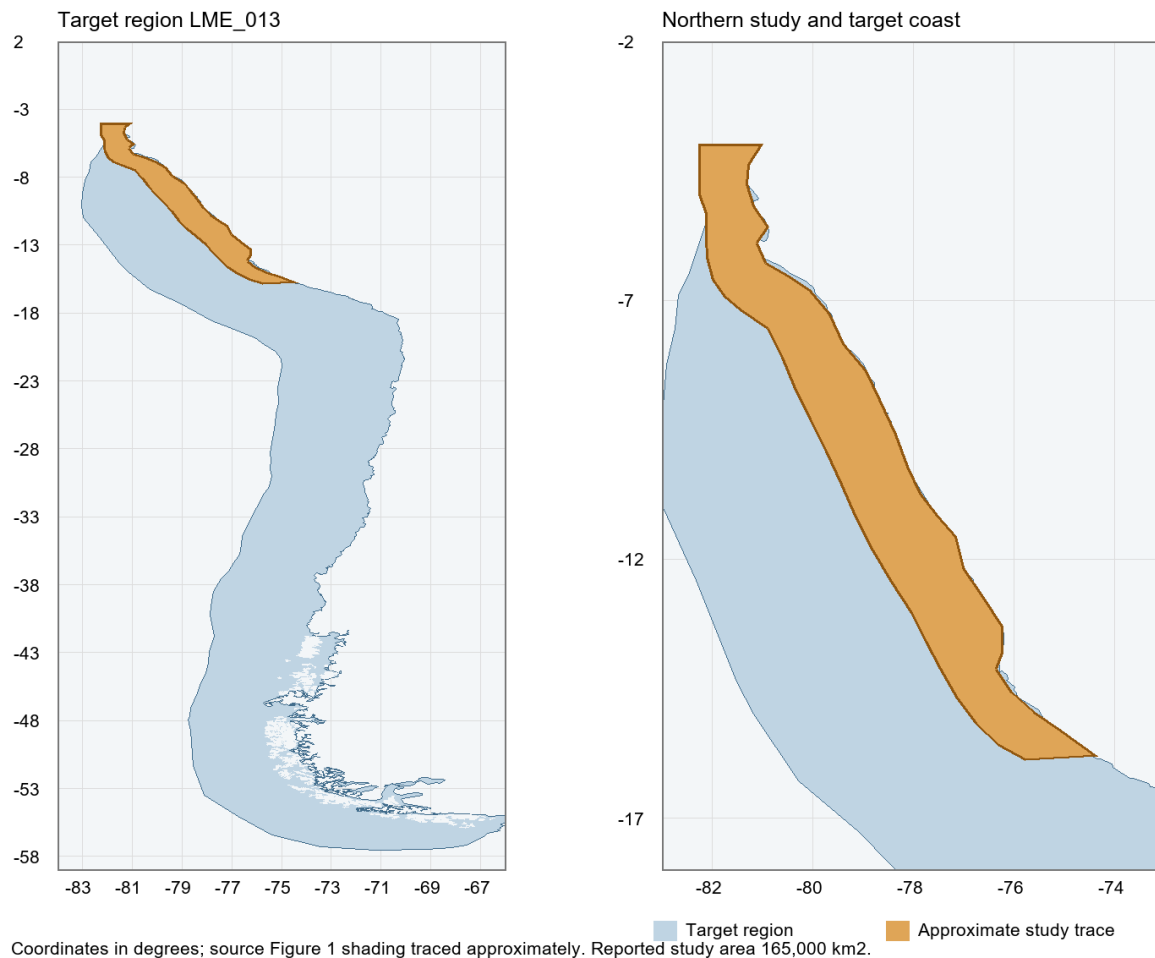
Geographic evidence

The study covers a northern coastal shelf segment of the target LME. The shaded source boundary was traced approximately against its graticule; reported area and traced area differ, so the A/B ranges express boundary uncertainty.

Region area

Target region boundary

Humboldt Current boundary and Northern Humboldt study



Humboldt Current LME_013 Sea Around Us boundary and approximate study trace in common geographic coordinates. [Boundary evidence](#)

Study area in the article

Study boundary for the selected model

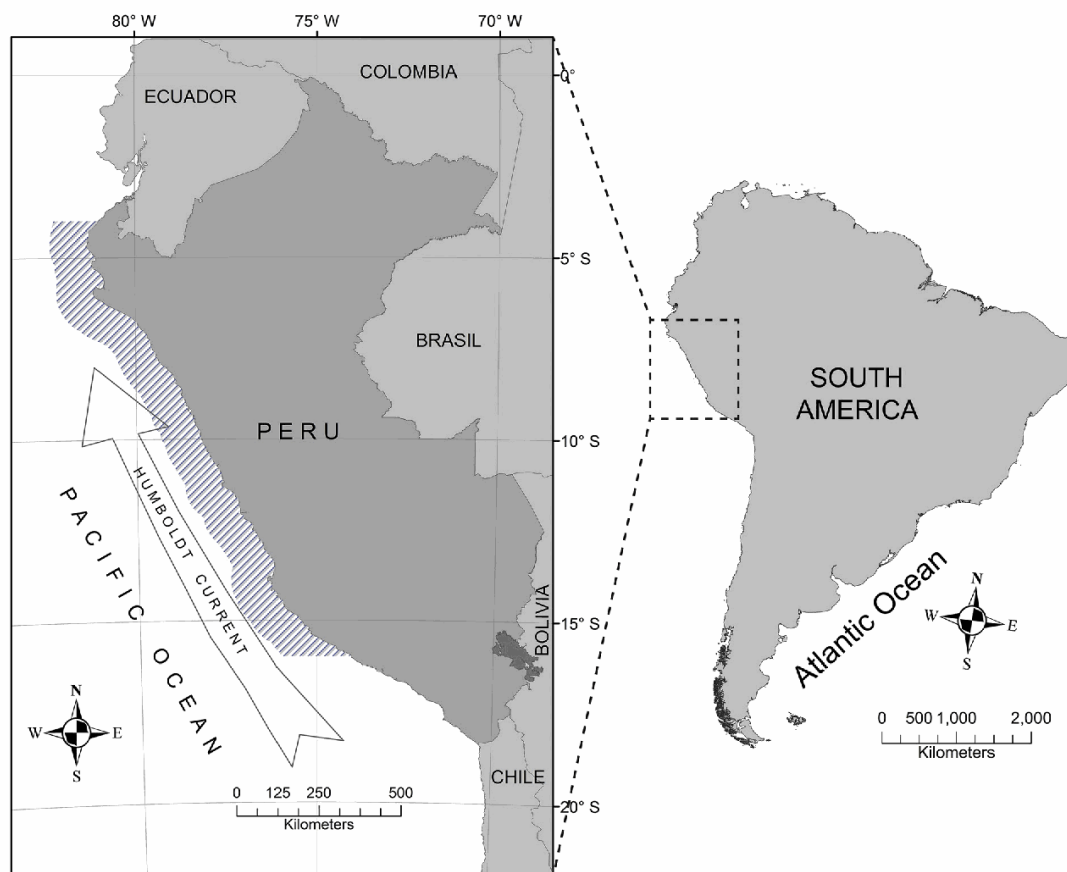


Fig. 1. Map of the study area (shaded area) covering from 4°S to 16°S, and up to 111 km (60 nm) from the coastline, in the Northern Humboldt Current Ecosystem.

Chiaverano et al. (2018), original Figure 1, PDF page 2 / printed page 29. Hatching marks the 4°S–16°S coastal domain, up to 111 km offshore. [Source PDF](#)